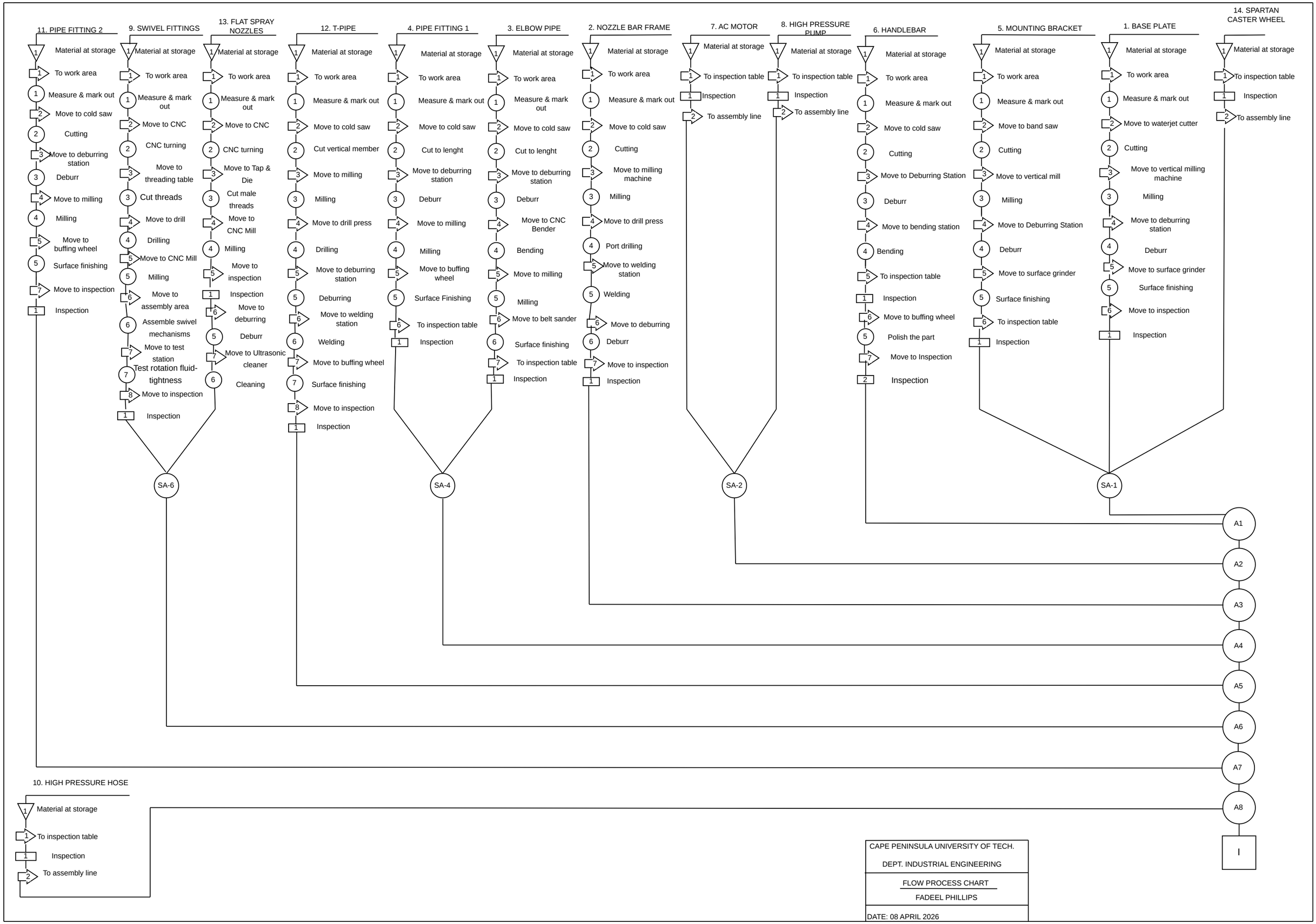

11. PIPE FITTING 2
Material at storage
To work area
Measure & mark out
Move to cold saw
Cutting
Move to deburring station
Deburr
Move to milling
Milling
Move to buffing wheel
Surface finishing
Move to inspection
Inspection
9. SWIVEL FITTINGS
Material at storage
To work area
Measure & mark out
Move to CNC
CNC turning
Move to threading table
Cut threads
Move to drill
Drilling
Move to CNC Mill
Milling
Move to assembly area
Assemble swivel mechanisms
Move to test station
Test rotation fluid-tightness
Move to inspection
Inspection
13. FLAT SPRAY NOZZLES
Material at storage
To work area
Measure & mark out
Move to CNC
CNC turning
Move to Tap & Die
Cut male threads
Move to CNC Mill
Milling
Move to inspection
Inspection
Move to deburring
Deburr
Move to Ultrasonic cleaner
Cleaning
12. T-PIPE
Material at storage
To work area
Measure & mark out
Move to cold saw
Cut vertical member
Move to milling
Milling
Move to drill press
Drilling
Move to deburring station
Deburring
Move to welding station
Welding
Move to buffing wheel
Surface finishing
Move to inspection
Inspection
4. PIPE FITTING 1
Material at storage
To work area
Measure & mark out
Move to cold saw
Cut to lenght
Move to deburring station
Deburr
Move to milling
Milling
Move to buffing wheel
Surface Finishing
To inspection table
Inspection
3. ELBOW PIPE
Material at storage
To work area
Measure & mark out
Move to cold saw
Cut to lenght
Move to deburring station
Deburr
Move to CNC Bender
Bending
Move to milling
Milling
Move to belt sander
Surface finishing
To inspection table
Inspection
2. NOZZLE BAR FRAME
Material at storage
To work area
Measure & mark out
Move to cold saw
Cutting
Move to milling machine
Milling
Move to drill press
Port drilling
Move to welding station
Welding
Move to deburring
Deburr
Move to inspection
Inspection
7. AC MOTOR
Material at storage
To inspection table
Inspection
To assembly line
8. HIGH PRESSURE PUMP
Material at storage
To inspection table
Inspection
To assembly line
6. HANDLEBAR
Material at storage
To work area
Measure & mark out
Move to cold saw
Cutting
Move to Deburring Station
Deburr
Move to bending station
Bending
To inspection table
Inspection
Move to buffing wheel
Polish the part
Move to Inspection
Inspection
5. MOUNTING BRACKET
Material at storage
To work area
Measure & mark out
Move to band saw
Cutting
Move to vertical mill
Milling
Move to Deburring Station
Deburr
Move to surface grinder
Surface finishing
To inspection table
Inspection
1. BASE PLATE
Material at storage
To work area
Measure & mark out
Move to waterjet cutter
Cutting
Move to vertical milling machine
Milling
Move to deburring station
Deburr
Move to surface grinder
Surface finishing
Move to inspection
Inspection
14. SPARTAN CASTER WHEEL
Material at storage
To inspection table
Inspection
To assembly line
SA-6
SA-4
SA-2
SA-1
A1
A2
A3
A4
A5
A6
A7
A8
I
10. HIGH PRESSURE HOSE
Material at storage
To inspection table
Inspection
To assembly line
CAPE PENINSULA UNIVERSITY OF TECH.
DEPT. INDUSTRIAL ENGINEERING
FLOW PROCESS CHART
FADEEL PHILLIPS
DATE: 08 APRIL 2026